KOdeCO net zero

CCS KNOWLEDGE SHARING WORKSHOP BY THE INNOVATION FUND

Aalborg, Denmark, 28th of November 2023

HOLCIM

DECARBONIZING KOROMAČNO CEMENT PLANT
ENABLING ISTRIA TO ALIGN INDUSTRY WITH TOURISM

35% reduction of CO_2 footprint

PHASE 1 - KOdeCO

Fossil fuel phase out

Minimize use of fossil fuel to 5% of energy needs

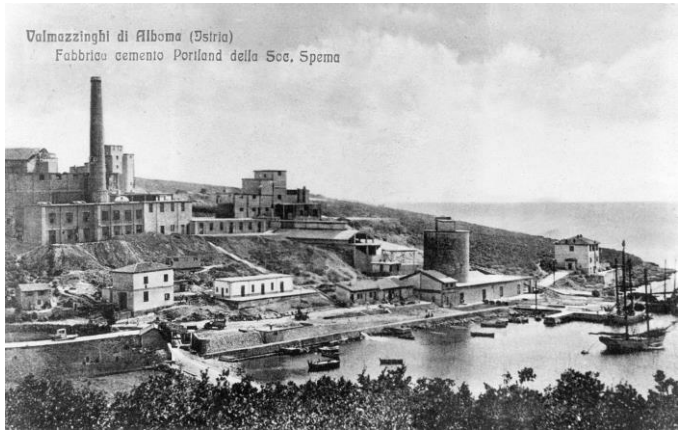
1926 — 1997 — 2022 — 2024 — 2025 — 2026 — 2028

PHASE 2 - KOdeCO net zero

Carbon Capture and Storage of all CO_2 emissions caused by the process

KOdeCO solar

Development of solar power plant



years of the Koromačno cement plant 1926 - 2026

„KOdeCO net zero” PROJECT DESCRIPTION
FIRST-OF-ITS-KIND, END-TO-END CCS VALUE CHAIN

Location: Koromačno, Croatia

Actual / projected captured CO2: 367,000 t CO2/year

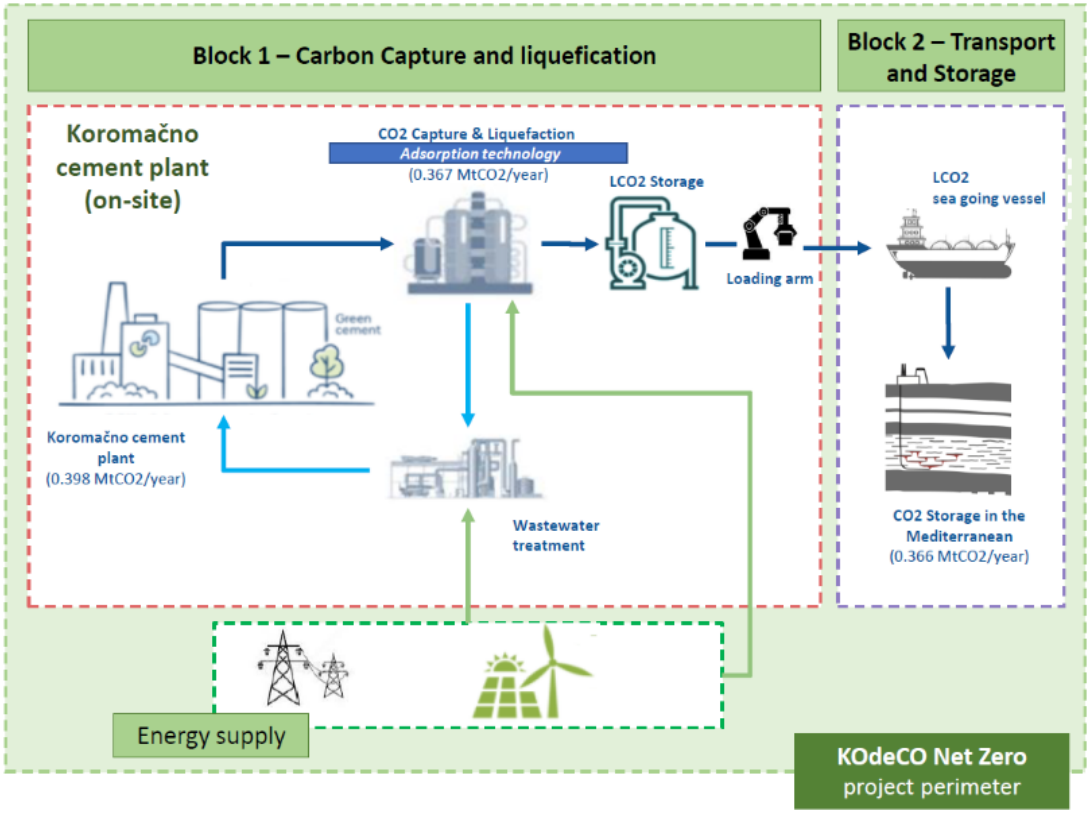
Technology: CO_2 capture adsorption technology

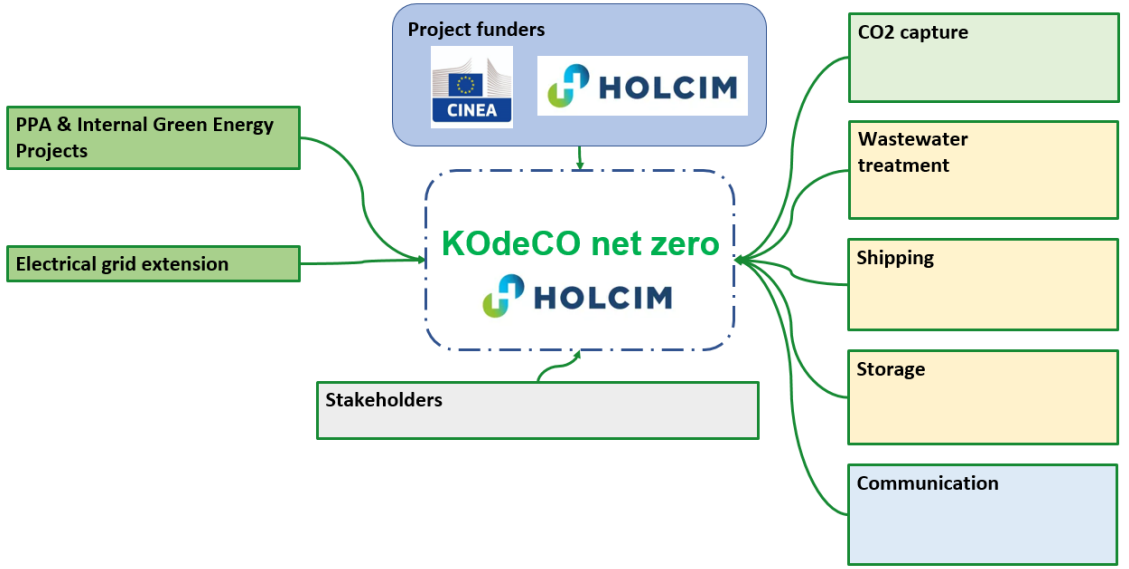
CO_2 Usage/storage model: Offshore sink

Estimated CAPEX (rough estimate): 237 mil. EUR

HOLCIM

SYNERGY WITH EXPERIENCED STRONG PARTNERS
RESULT IS INNOVATIVE AND SCALABLE PROJECT WITH HIGH MATURITY





PUBLIC FUNDING	EU InnovFund
	JTF
PRIVATE FUNDING	Holcim Group

Q3 2023
EU IF Results

Application to
EU InnovFund-2022-LSC

Q4 2025
Financial Closure

- FEED and basic engineering
- EIA
- Contract negotiation and finalization
- Permitting and compliance
- Preparation for FID

- Site preparation and energy delivery preparation
- EPC execution
- Transportation and storage contract monitoring
- Integration, operational and maintenance plans
- Commissioning

Q4 2028
Entry into operations

Operations; reporting and continuation of knowledge sharing

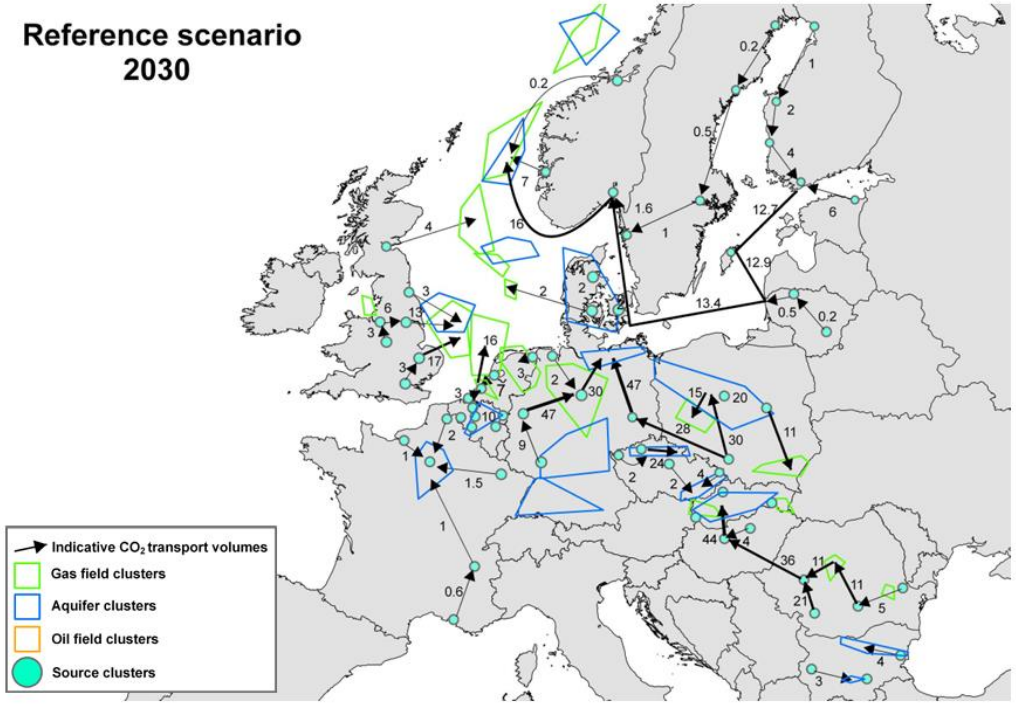
HOLCIM

JUST TRANSITION TERRITORY
WE ARE PROUD TO SUPPORT OUR COMMUNITY IN THE TRANSITION

Istria and **Sisačko-Moslovačka** counties are only territories in Croatia identified by Just Transition Fund (JTF) and included in Croatian Integrated Territorial Program

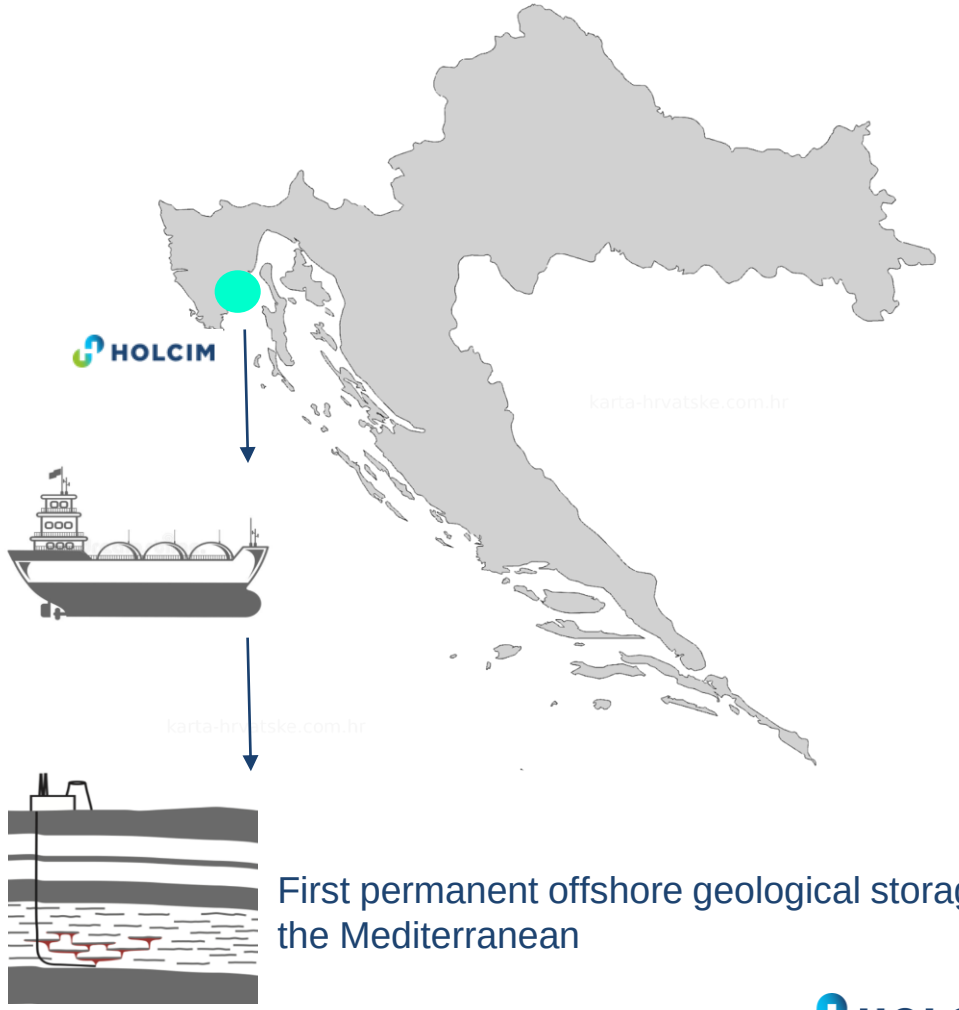
NORTH EUROPE CREATED OPPORTUNITIES
WE CARE ENABLING SOLUTIONS FOR INDUSTRY IN SOUTH EUROPE

TRANSPORTATION AND STORAGE NORTH - EU COUNTRIES



Source: CO2Europipe; WP2.2 Report – Development of a large-scale CO2 transport infrastructure in Europe:matching captred volumes ans storage availability

TRANSPORTATION AND STORAGE - KOdeCO net zero



First permanent offshore geological storage in the Mediterranean

HOLCIM

Project IFESTOS

Scaling Carbon Capture & Storage in Greece

CCS Knowledge Sharing workshop, EU Innovation Fund

Aalborg, November 28, 2023

Aris Tsikouras, Titan Cement

Value chain

Overview

Titan perimeter

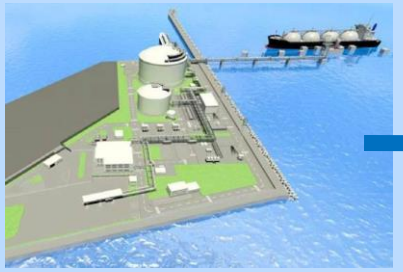
Carbon Capture
Kamari plant
1.9 mtpy CO_2

TITAN

- Pre-combustion
- Post-combustion
- Compression

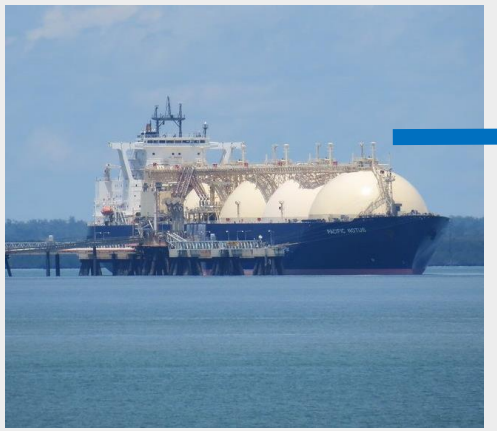
Other emitters

Industrial Hub
Elefsina

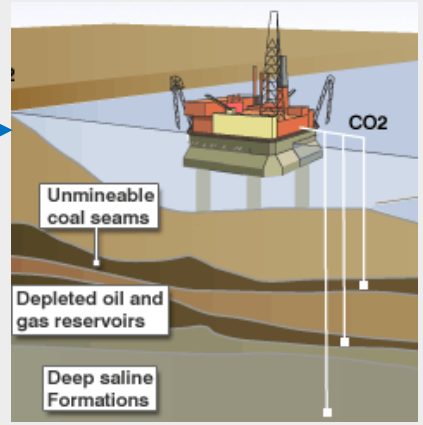


- CO_2 Pipeline
- Liquefaction unit
- Buffer storage & loading terminal

Sea Transport
via liquid CO_2 vessels



Permanent Geological Storage
Prinos or other suitable site in the Med



Expected entry into operation: 2029

Regulatory challenges

- ✓ Provide open & non-discriminatory access to logistics & storage infrastructure
- ✓ Ensure unbundled, cost-based, transparent tariffs
- ✓ Establish Carbon Contracts for Difference
- ✓ Determine licensing & certification framework
- ✓ Define technical / safety standards
- ✓ Enable cross-border transportation

iFESTOS
Thank you

IRIS

Innovative low carbon hydrogen and methanol production by large scale carbon capture

CCS Knowledge Sharing workshop by the Innovation Fund – Realising opportunities along the value chain

27-28 November 2023

MOTOR OIL

IRIS | Innovation Fund

IRIS project **comprises a novel, heavily integrated point-source CCUS technology solution**, applied on its current hydrogen production process and its coupling to a small-scale methanol production unit, which will utilize part of the captured CO_2 as feedstock.
MOH's innovative project IRIS, seeks to couple **the production of ultra-low carbon hydrogen and methanol by integrating point-source carbon capture**, applied on its current Steam Methane Reforming unit, electrolytic H_2 production and a catalytic process of high selectivity.

Coordinator: Motor Oil (Hellas) Corinth Refineries S.A.

Location: Agioi Theodoroi, Corinth, Greece

Sector: Refineries

Amount of IF grant: €126.790.000

Duration: 168 months

Volume of CO_2 captured for storage by 2030 (Mtpa): 1.2Mt

Starting date – Ending date: January 2024 – June 2037

Planned date of entry into operation: July 2028

MOH Refinery, Agioi Theodoroi, Corinth, Greece

MOTOR OIL

IRIS | Key project aspects

IRIS constitutes a significant step:

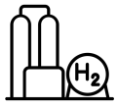
(a) in the decarbonization plan of MOH carbon capture project implemented in the Steam Methane Reforming Unit of the refinery with post-combustion capture technology scheme which allows maximization of the CO_2 captured

- decreases the total carbon footprint of the refinery by 25% (~495,000 t/y of CO_2 captured)
- achieves very high energy integration with the existing refinery operations, further reducing its total energy requirements
- will lay the foundation for the future expansion of the CO2 capture technology to two more refinery units, that could further decrease the total refinery carbon footprint by 50%

(b) for the production of low CO_2 energy carriers for mobility purposes and other industrial usage

- 55,280 t/y of blue hydrogen produced - ~37% lower than the EU Taxonomy threshold
- 10.000 t/y e- methanol as part of the captured CO_2 will be combined with green hydrogen produced by a 30MW electrolyzer (EPHYRA), materializing the principles of circular economy

(c) to lever the development of the very few CO_2 storage facilities in the Mediterranean Sea



(d) for the creation of a wider ecosystem of projects that promote the development of hydrogen market, namely the EU-backed projects EPHYRA* and TRIERES*

MOTOR OIL

* EPHYRA and TRIERES projects are supported by the Clean Hydrogen Partnership and its members Hydrogen Europe and Hydrogen Europe Research under Grant Agreements No. 101112220 & No. 101112056 respectively

Co-funded by the European Union Clean Hydrogen Partnership

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Project OLYMPUS

Milaki Cement Plant Decarbonization

A MEMBER OF HOLCIM GROUP

HIRACLES GROUP OF COMPANIES

Milaki Cement Plant

Operating in Evia

Since 1982

Capacities

Clinker : 1,600 kt/a
Cement : 1,700 kt/a

2 Owned Quarries

Limestone & Schist
50 Years Reserves

2 Alternative Fuel Workshops

for Biomass
and SRF

Bagging Facilities

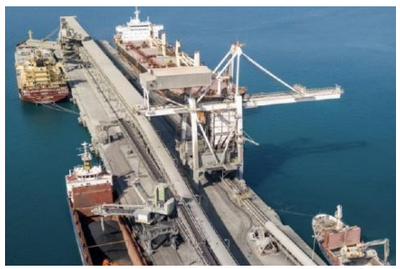
Ground Cement, Solid Fuels
and Raw Solid Fuels

Bulk Material Yard

400kt Storage
Capacity

Innovation & Differentiation

Cement & Clinker



- Cement exports worldwide
- 24/7 simultaneous loading & unloading ability
- Up to 150 kt dead weight capacity

Oil Well Cement

- API certified manufacturer of Well Cement for the Oil & Gas Industry (Class G)
- Deliver Oil Well Cement to end customers (Europe, Middle East, Africa, Oceania)
- Bagging facilities specifically designed to meet O&G industry needs

Solid Fuels

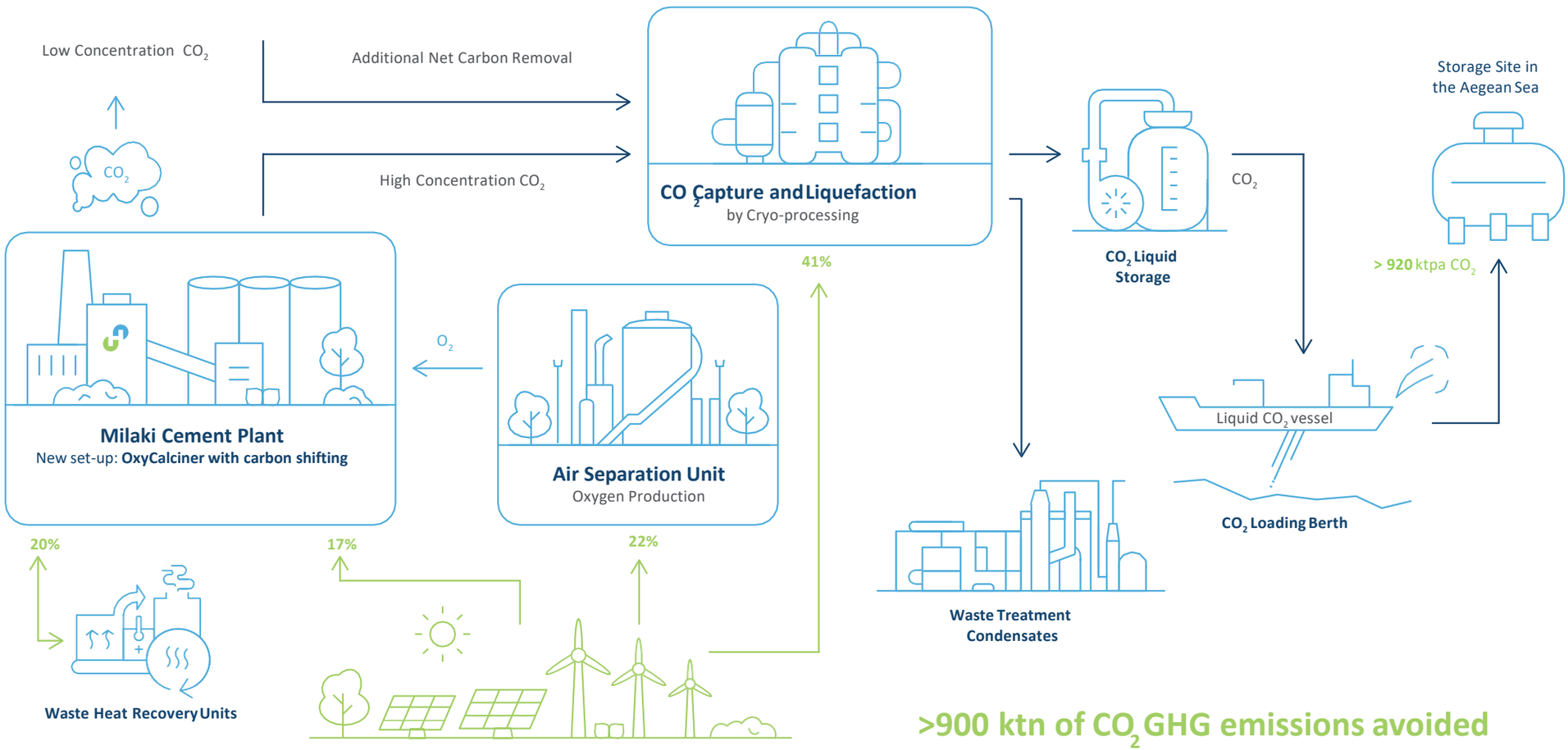
- Trading of tailor-made SF satisfying customer requirements (ability for a variety of processes e.g. grinding, screening, bagging, mixing)
- Increased flexibility to manage different fuel types for customers

Aggregates

- Sales to greek islands through plant port

OLYMPUS Project

2028 Milaki Cement Plant Carbon Capture & Storage



>900 ktn of CO_2 GHG emissions avoided

Main benefits

1

Greece enacting a leadership role in **climate action**, turning the latter to an **opportunity for growth** and **moving the needle on sustainability**.

2

Establishing Greece as a friendly operating environment for forward- looking enterprises; thus, reinforcing its **investment potential**.

3

Not only safeguarding current **job positions** but also **creating new** and more specialized ones, especially in regions where underemployment prevails, such as Central Greece.

4

Kick-starting the technology race to a **CO_2 circular economy & net zero transition** in the cement sector across Europe with **Greece** being **in the spotlight**.

5

Serving as a blueprint for the conversion of cement plants both for Greece and the whole Europe towards a sustainable constructions sector with multiple benefits to the **environment** and the **society**, but also to the **profitability** and **competitiveness** of the Greek and EU cement industry.

6

Contributing to the significant **reduction of GHG emissions through net carbon removal** and paving the way for **green growth**, not limited to cement sector, but for the rest of CO_2 intensive industry as well.

7

CO_2 avoidance represents 2% of total Greece's CO_2 emissions

HIRACLES
GROUP OF COMPANIES

BECCS at Växjö Energi, Sweden

Julia Ahlrot, Project leader Växjö Energi AB julia.ahlrot@veab.se
Daniel Eidenskog, Senior Process Engineer Växjö Energi AB, daniel.eidenskog@veab.se

Växjö Energi, a public owned energy company

Founded
1887

Employees
222

Owner
Växjö kommun

Power Grid

Number of customers

37 700

Total grid
152.000 km

Broadband

Number of customers

43 000

Total grid
435.000 km

Combined Heat&Power

Number of customers

10 100

Total grid
46.000 km

Växjö, a municipality in southern Sweden

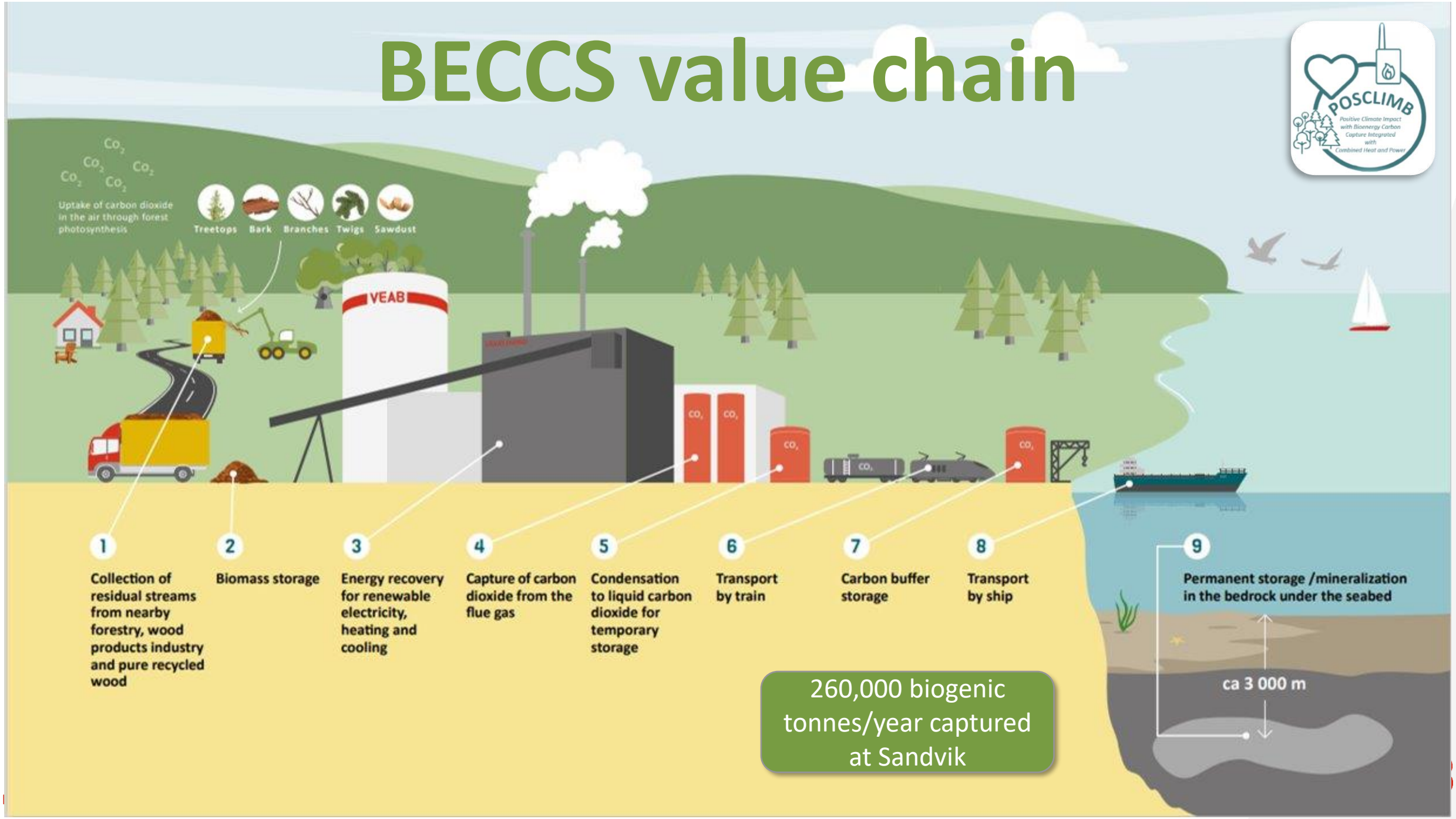
VEAB
VÄXJÖ ENERGI

EXISTING COMBINED HEAT AND POWER AT THE SANDVIK PLANT

100 % Fossil fuel free!
Biomass from forest residues

VEAB
VÄXJÖ ENERGI

BECCS value chain



260,000 biogenic tonnes/year captured at Sandvik

CLUSTER COOPERATION IN SOUTHERN SWEDEN FOR CARBON LOGISTICS

Subsidied by
Swedish Energy Agency
Finansieras av Europeiska unionen
NextGenerationEU

CNetSS

CARBON NETWORK SOUTH SWEDEN

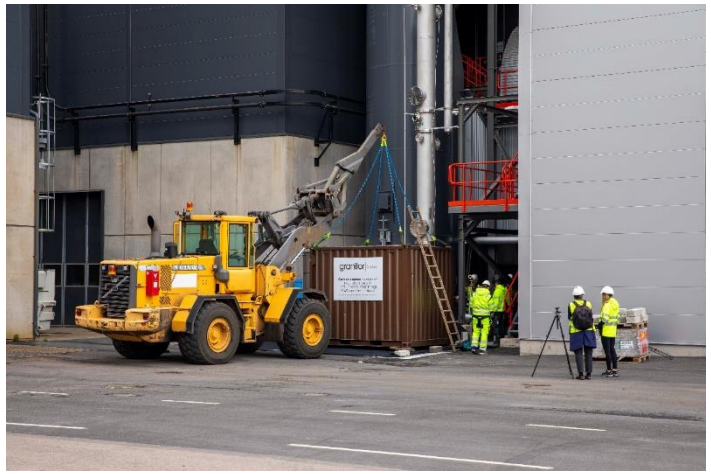
VEAB VÄXJÖ ENERGI
kraftringen
NORDION ENERGI
CMP COPENHAGEN MALMÖ PORT
SYSAV
stora enso
ÖRESUNDS KRAFT
EN DEL AV DIN VARDAG
e·on
Kemira
Höganäs
VEAB VÄXJÖ ENERGI

PILOT PROJECT FOR CARBON CAPTURE IN COOPERATION WITH LUND UNIVERSITY

Subsidied by

Swedish Energy Agency

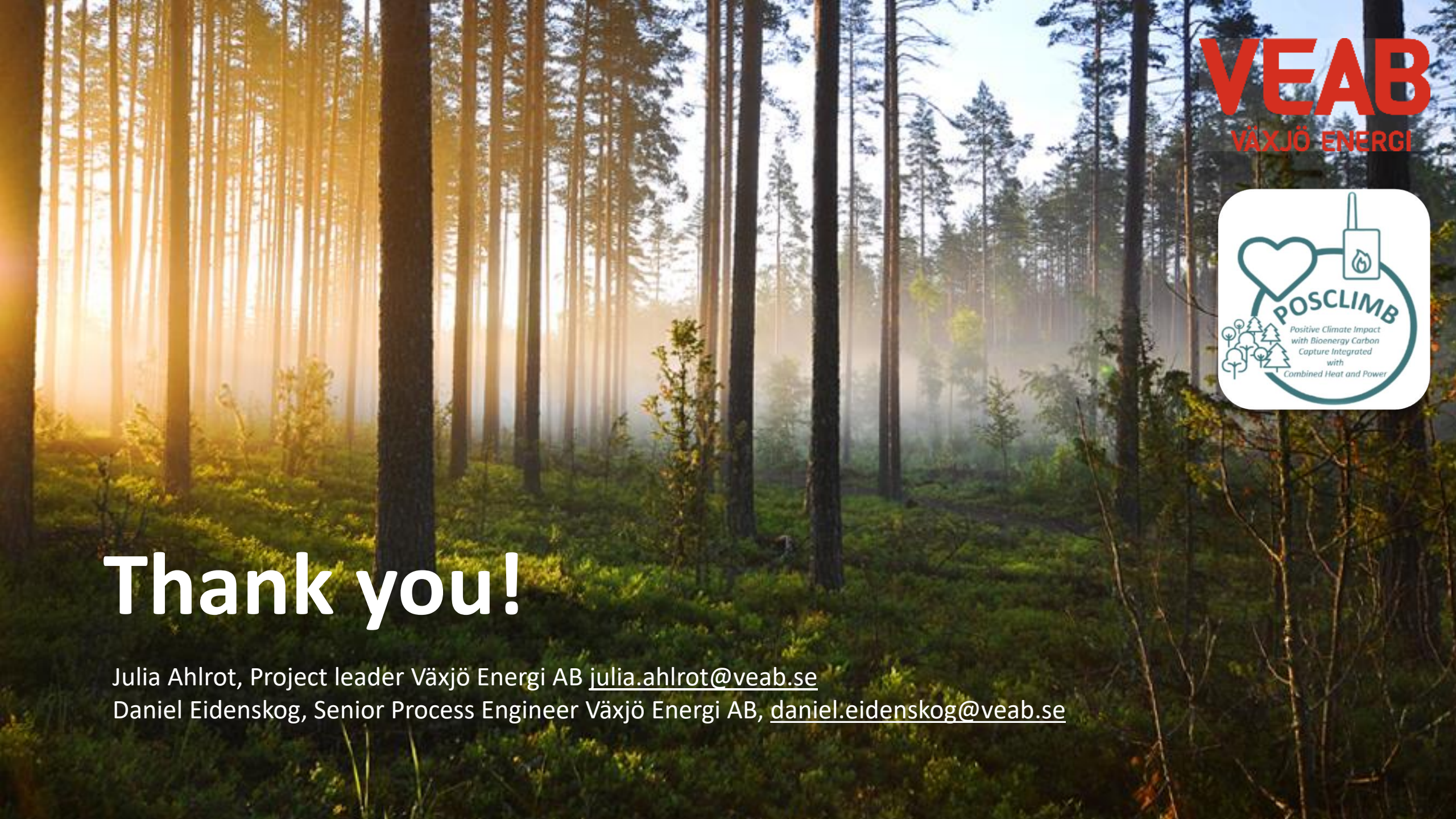
Finansieras av Europeiska unionen
NextGenerationEU



Amine technology
AMP/DMSO

EN DEL AV DIN VARDAG

LUNDS UNIVERSITET
LTH LUNDS TEKNISKA HÖGSKOLA
granitor
SYSAV
ÖRESUNDS KRAFT
VEAB VÄXJÖ ENERGI



Thank you!

Julia Ahlrot, Project leader Växjö Energi AB julia.ahlrot@veab.se
Daniel Eidenskog, Senior Process Engineer Växjö Energi AB, daniel.eidenskog@veab.se

CO2NTESSA PROJECT

PROJECT BACKGROUND

PROJECT BACKGROUND

Cement plant NEXE in Našice accounts for 1/3 of cement production in Croatia - more than 1 million t of cement per year, out of which more than 50% of production is exported to the countries of the region.

PROJECT BACKGROUND

Cement production represents energy-intensive process with large amount of CO2 emission as by-product. More than 70% of CO2 emissions is a result of chemical process of decomposition of carbonate raw material at >1400°C and Carbon Capture technology represents the only long-term solution for reduction of CO2 emissions.

The production of 1 million tons of cement requires >900 GWh of energy

CO_2NTESSA PROJECT

CO_2NTESSA PROJECT

Project: CO_2NTESSA

Industry: Cement

Location: Našice/Slavonia/Croatia

Investment: EUR 391.7 mil

Entry into operation: 2029

Category: Carbon Capture and Storage (CCS)

Technology: Thyssenkrupp Polysius® pureoxyfuel Carbon Capture

March 2023: Applied on InnovFund-2022-LSC-01-GENERAL

July 2023: The European Commission (CINEA) proposed to the European Investment Bank (EIB) that, for the CO_2NTESSA project, the allocation of EIB technical assistance (EIB PDA) should be considered.

October 2023: The CO_2NTESSA project is included in the list of strategic investment projects of the Republic of Croatia.

CO_2NTESSA PROJECT

The cement industry is very energy intensive and one of the most significant emitters of CO_2, responsible for around 4% of all emissions at the EU level. The need to decarbonise cement industry is urgent to prevent the risk of locking in CO_2 emissions beyond 2050, when the EU plans to achieve full carbon neutrality.

The CO_2NTESSA project will lead to implementation of an innovative carbon capture technology at NEXE cement plant in Našice, Croatia. Existing plant will be modified based on the Polysius PureOxyfuel technology developed by thyssenkrupp. **Based on this, the CO_2NTESSA project will allow capture 739 000 t CO_2/year, making cement production at NEXE close to zero emissions. Moreover, the CO_2NTESSA project will unlock the potential for NEXE to become the first negative emitter of CO_2 in the EU because of use of alternative fuels.**

Geologically highly appropriate, onshore storage location Bockovci is located only 38 km away, allowing energy and cost-efficient injection of the captured CO_2 into the saline aquifer. The CO_2NTESSA project will achieve synergy with the GT CCS project which includes renewal of the existing out-of-function pipeline for transport of captured CO_2 from the NEXE cement plant. The GT CCS project is expected to be included into the EU's list of Projects of Common Interest (PCI), with potential to become a regional hub for CO_2, making the CO_2NTESSA project an important milestone of carbon capture and storage development in Croatia and beyond.

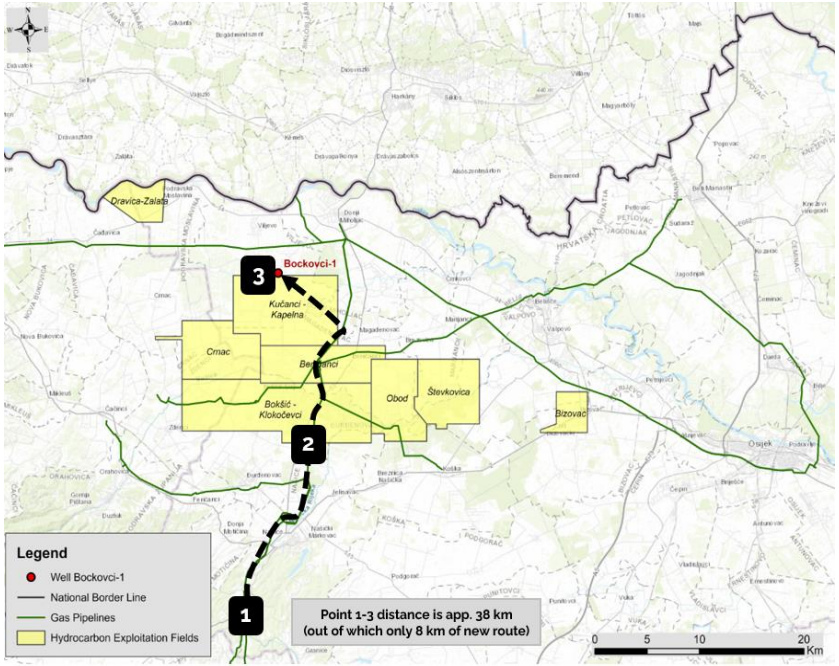
Carbon capture technologies in cement production require massive investments, making high costs the main barrier to wide roll out of carbon capture solutions, so public funding is crucial in reaching energy and climate objectives within the cement industry. **Total investment cost in the CO_2NTESSA project will be EUR 391.7 million, making it one of the largest planned investments in industry in Croatia**. Project CO_2NTESSA enables capturing of CO_2 in more cost-effective way in comparison with similar Carbon Capture projects, securing the long-term competitiveness of cement products abroad.

INNOVATIVENESS OF THE PROJECT

INNOVATIVENESS OF THE PROJECT

A unique, fully rounded, cost-effective CCS project in Southeast Europe. One of few in the EU that has efficient solution for disposal of captured CO_2 by transport pipeline up to location Bockovci-1, where CO_2 will be injected into the deposit-saline aquifer (connection to the project Croatia GT CCS).

1.	CO_2NTESSA PROJECT	NEXE	CO_2 CAPTURE
2.	GT CCS CROATIA	PLINACRO	TRANSPORT OF CO_2
3.	GT CCS CROATIA	CROATIAN HYDROCARBON AGENCY	STORAGE OF CO_2



INNOVATIVENESS OF THE PROJECT

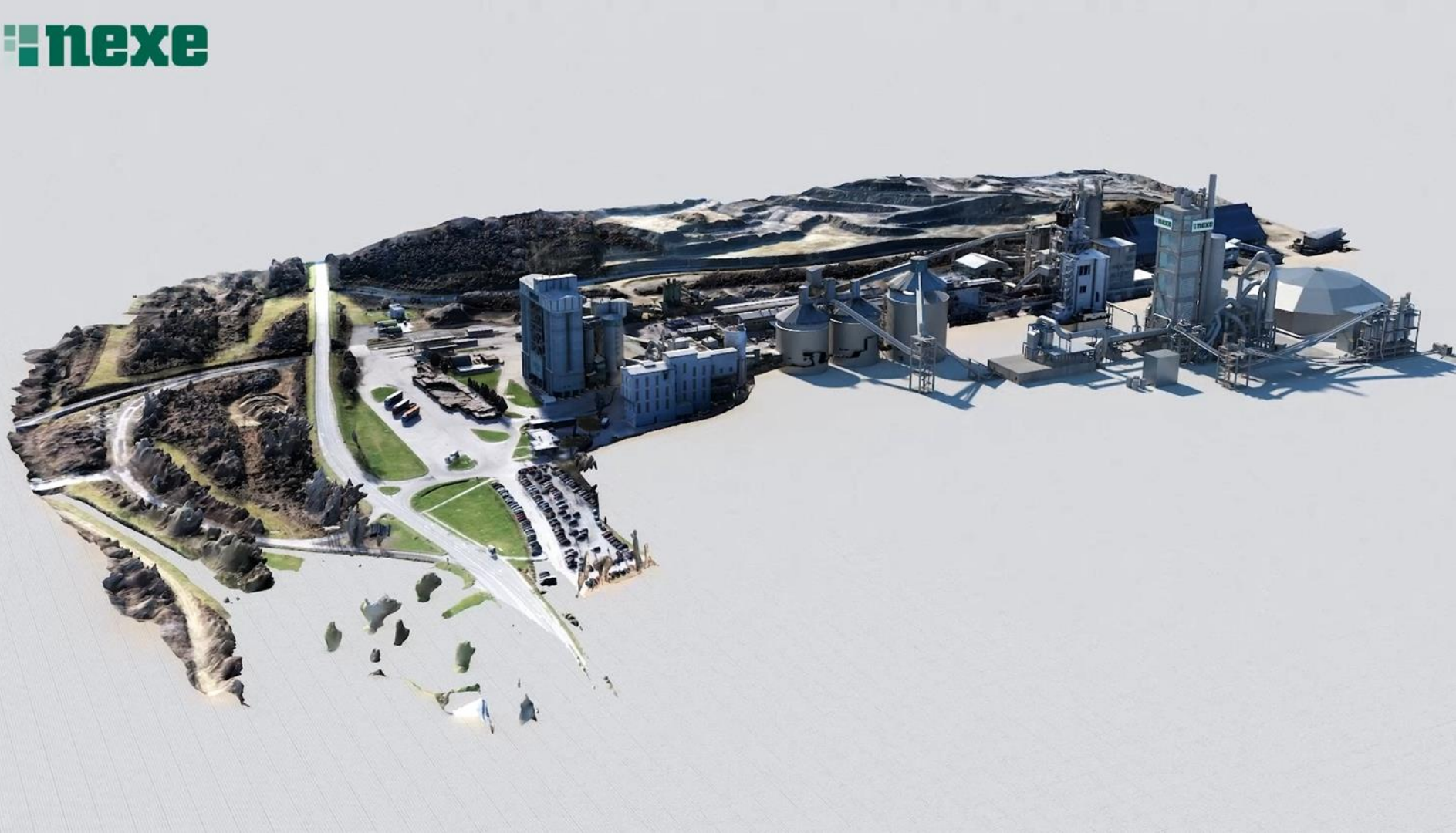
Carbon Capture industrial large-scale project which do not need additional energy-intensive CO_2 purification for further transport and injection/storage of CO_2, as CO_2 purity of app 90% is enough.

Unique completely rounded CCS project in south-east Europe, that includes CO_2 capture, transport of CO_2 by pipeline and injection of CO_2 in saline aquifer Bockovci-1 (on-shore storage) which is only 38 km far from cement plant.

Cost efficiency through significant less CAPEX and OPEX (per tonne of CO_2 avoided) in comparison with similar projects.

NEXE cement plant in Našice becomes a negative CO_2 emitter with capture of >700.000t CO_2 per year (100% CO_2 emissions from production process) and use of alternative fuels (enables continued substitution of alternative fuels with a substitution coefficient >90%).

Implementation of carbon-neutral and cost-efficient production as guideline for other plants in industry that have low quality raw materials and low level of CO_2 concentration in exhaust gases from production processes (cement plants, power plants, refineries...) as an example of green transition for long-term sustainable operations.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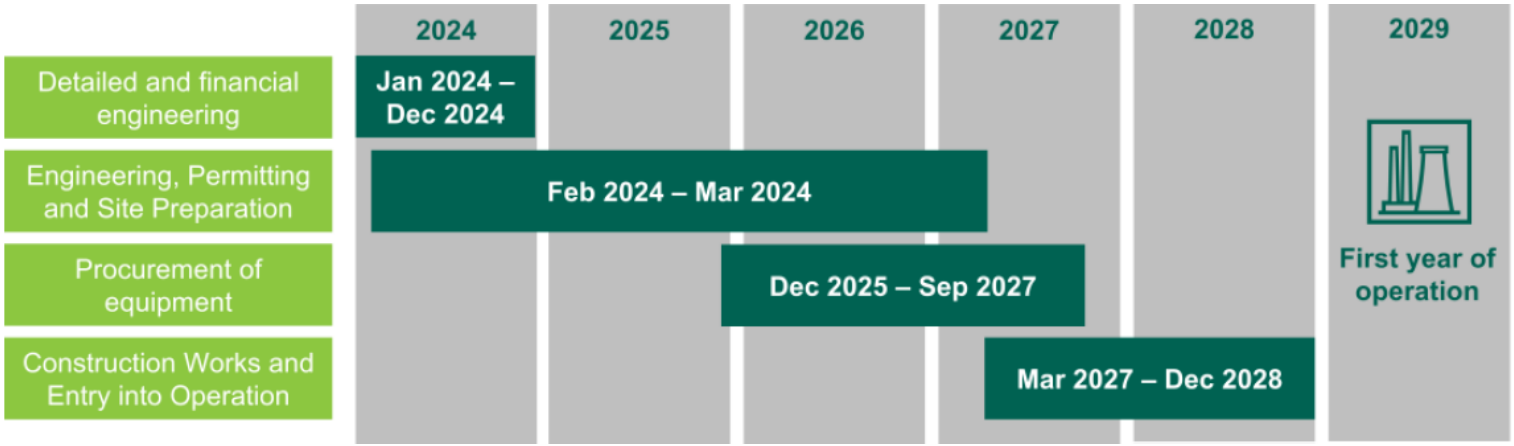
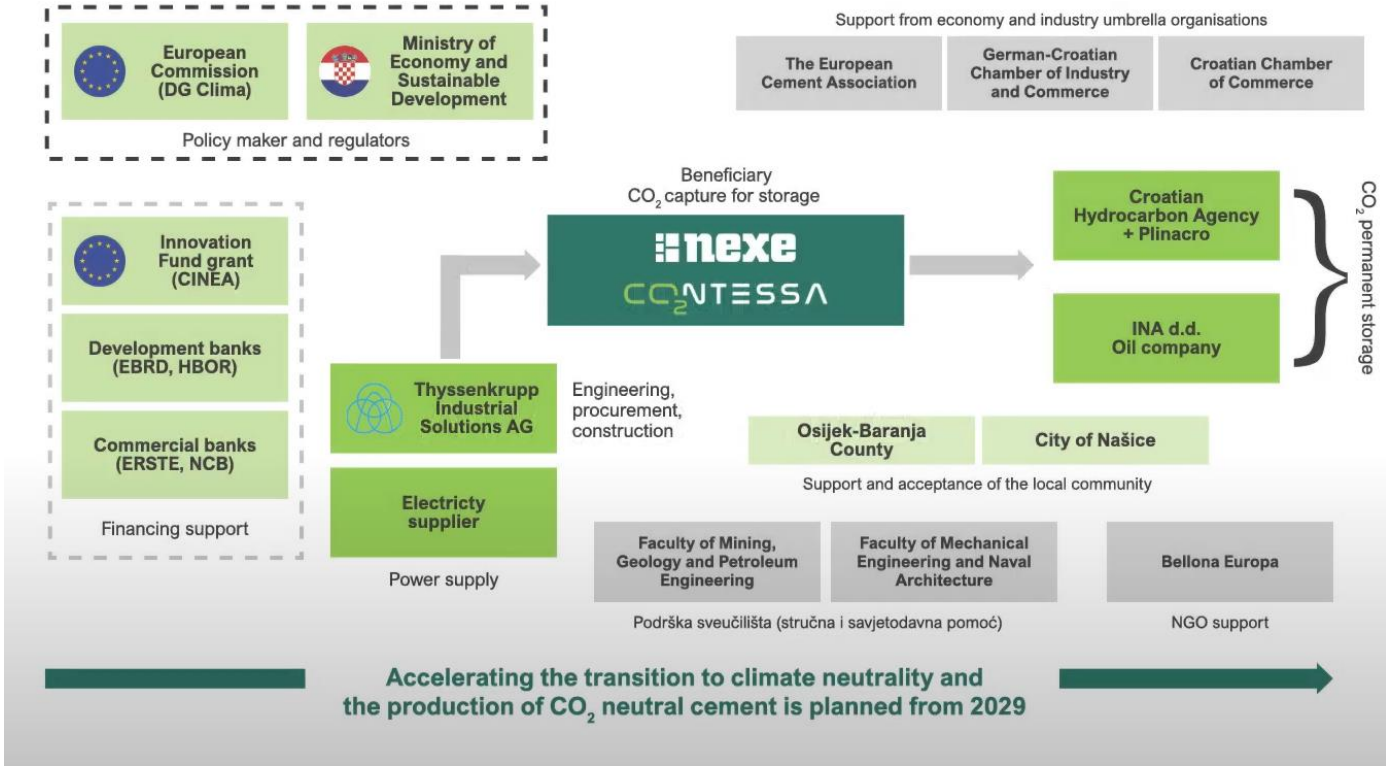
BOCKOVCI

PROJECT IMPLEMENTATION

PROJECT IMPLEMENTATION



INFORMATION ON PROJECT APPLICATION TO THE INNOVATION FUND



CO2NTESSA

More information available on:

www.nexe.hr/en/co2ntessa/

nexe